Integer Linear Programming

Further Problems

Mixed

All integer

Binary

Sensitivity: Internal

Minimum Set Cover

Frustration

Sensitivity: Internal

General Set Covering Problem

minimise

$$\sum_{i=1}^{N} x_i$$

subject to:

$$\sum_{i:e \in s_i} x_i \geq 1, \qquad \forall e \in U$$

$$x_i \in \{0,1\}$$

S – All possible sets to choose from

$s_i \in S$ – A given set s_i

x_i - Set s_i chosen or not

N – number of sets |S|

U = Set of all elements contained in S

Sensitivity: Internal

General Set Covering Problem



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subject to:

$$\sum_{i:e\in s_i} x_i \geq 1\,, \qquad \forall e \in U$$

$$x_i \in \{0,1\}$$

Example

$\{1,2,3,4,5\}$

$s_1 = \{1,2,5\}, \quad s_2 = \{2,3\},$
$s_3 = \{4,5\}, \quad s_4 = \{3,4\}$

$S = \{s_1, s_2, s_3, s_4\}$
$U = \{1,2,3,4,5\}$ ← elements
$N = 4$

Pick the minimum number of sets to get all elements in U

Answer by inspection?

2 sets

$x_1 = 1$, $x_2 = 0$
$x_3 = 0$, $x_4 = 1$

Sensitivity: Internal

General Set Covering Problem

$x_i = \begin{cases} 1 & \text{Pick } s_i \\ 0 & \text{Don't pick } s_i \end{cases}$

minimise

$$\sum_{i=1}^{N} x_i$$

subject to:

$$\sum_{i:e \in s_i} x_i \geq 1, \qquad \forall e \in U$$

$$x_i \in \{0,1\}$$

$s_1 = \{1,2,5\}, \quad s_2 = \{2,3\}, \quad s_3 = \{4,5\}, \quad s_4 = \{3,4\}$

$S = \{s_1, s_2, s_3, s_4\}$
$U = \{1,2,3,4,5\}$
$N = 4$

minimise

$x_1 + x_2 + x_3 + x_4$ (1 0 0 1) $= 2$

subject to:

$x_1 \geq 1$	(Element 1)
$x_1 + x_2 \geq 1$	(Element 2)
$x_2 + x_4 \geq 1$	(Element 3)
$x_3 + x_4 \geq 1$	(Element 4)
$x_1 + x_5 \geq 1$	(Element 5)

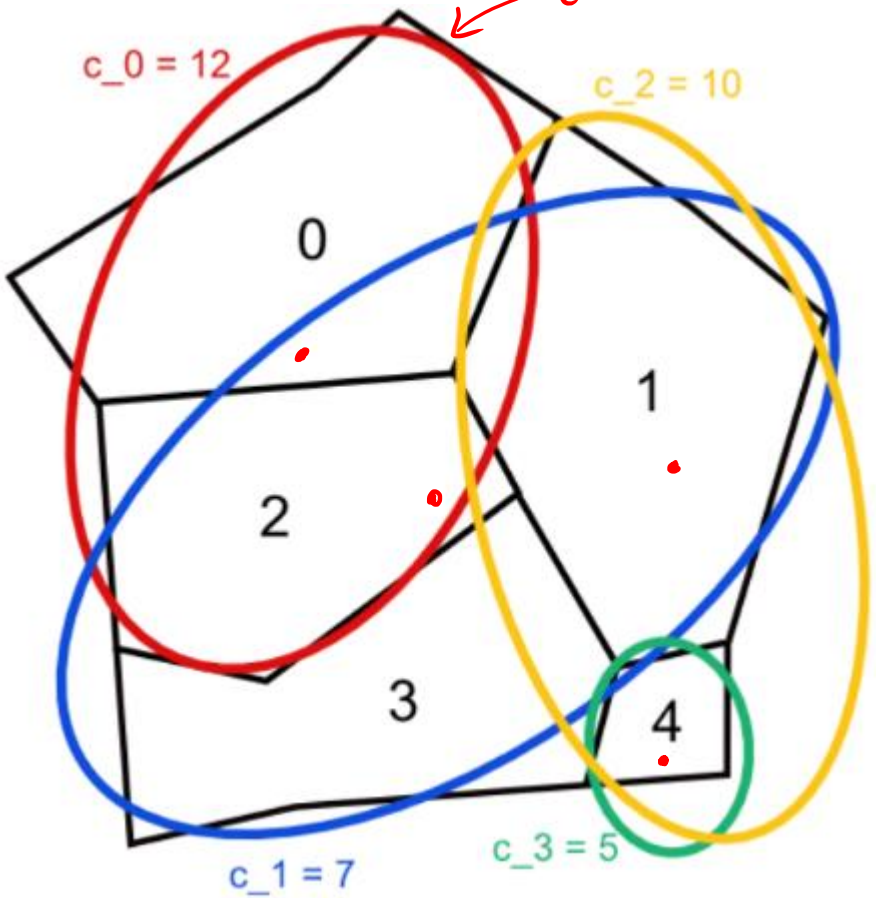
$x_s \in \{0,1\}$

Minimum set cover 2

$\{s_1, s_4\}$

Sensitivity: Internal

Costed Firestation example



cover of a firestation

S – All possible firestations to choose from
$s_i \in S$ – A given firestation s_i
x_i - firestation s_i chosen or not
c_i - cost of running firestation i
U = Set of all cities contained in S

$S = \{s_0, s_1, s_2, s_3\}$ ← Firestations
$U = \{0,1,2,3,4\}$ ← cities / regions
$N = 4$ ← no of firemen.

$x_0 \in \{0,1\}$ ...
x_1 ...
x_2 ...
x_3 ...

$s_0 = \{0,2\}$
$s_1 = \{1,2,3\}$
$s_2 = \{1,4\}$
$s_3 = \{4\}$

Firestations

Vodafone shops

$c_0 = 12$
$c_1 = 7$
$c_2 = 10$
$c_3 = 5$

← cost firestations

Sensitivity: Internal

Costed Firestation example

S – All possible firestations to choose from
$s_i \in S$ – A given firestation s_i
x_i - firestation s_i chosen or not
c_i - cost of running firestation i
U = Set of all cities contained in S

$$S = \{s_0, s_1, s_2, s_3\}$$
$$U = \{0,1,2,3,4\}$$

$s_0 = \{0,2\}$
$s_1 = \{1,2,3\}$
$s_2 = \{1,4\}$
$s_3 = \{4\}$

$c_0 = 12$
$c_1 = 7$
$c_2 = 10$
$c_3 = 5$

minimise

$$\sum_{i=1}^{N} c_i x_i$$

subject to:

$$\sum_{i:e \in s_i} x_i \geq 1, \quad \forall e \in U$$

$$x_i \in \{0,1\}$$

minimise

$$c_0 x_0 + c_1 x_1 + c_2 x_2 + c_3 x_3$$

subject to:

$x_0 \geq 1$ (City 0)
$x_1 + x_2 \geq 1$ (City 1)
$x_0 + x_1 \geq 1$ (City 2)
$x_1 \geq 1$ (City 3)
$x_2 + x_3 \geq 1$ (City 4)
$x_s \in \{0,1\}$

Sensitivity: Internal

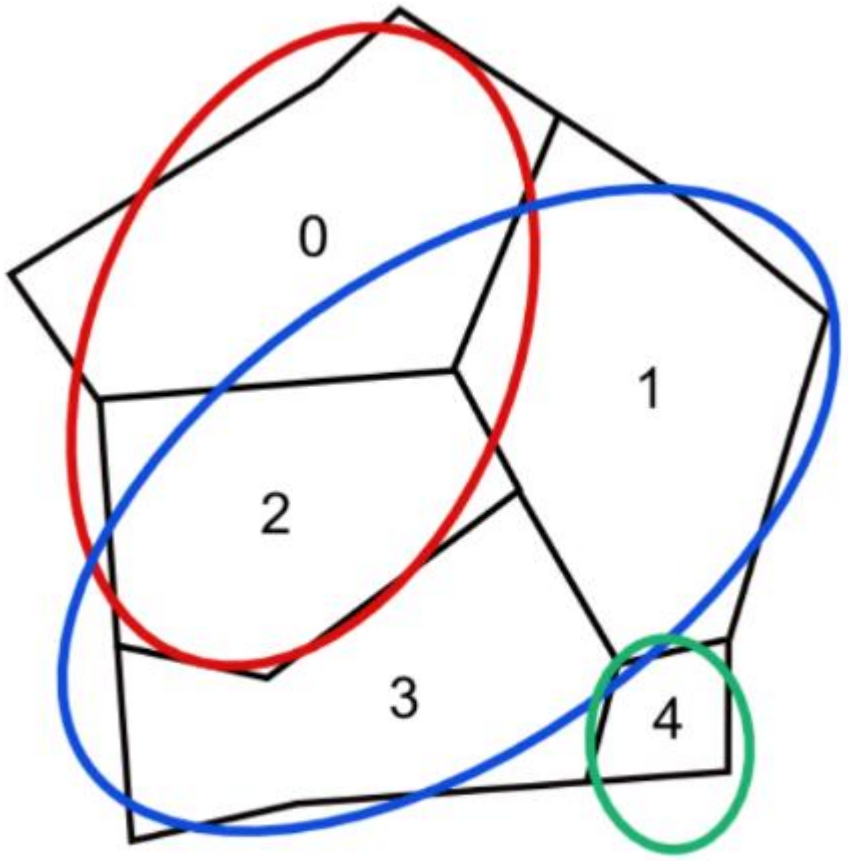
Costed Firestation example

S – All possible firestations to choose from
$s_i \in S$ – A given firestation s_i
x_i - firestation s_i chosen or not
c_i - cost of running firestation i
U = Set of all cities contained in S

$$S = \{s_0, s_1, s_2, s_3\}$$
$$U = \{0,1,2,3,4\}$$

$$s_0 = \{0,2\}$$
$$s_1 = \{1,2,3\}$$
$$s_2 = \{1,4\}$$
$$s_3 = \{4\}$$

$$c_0 = 12$$
$$c_1 = 7$$
$$c_2 = 10$$
$$c_3 = 5$$



cost = 12 + 7 + 5 = 24

Sensitivity: Internal

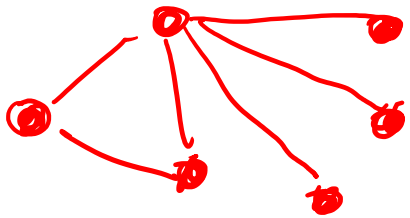
Minimum Vertex Cover

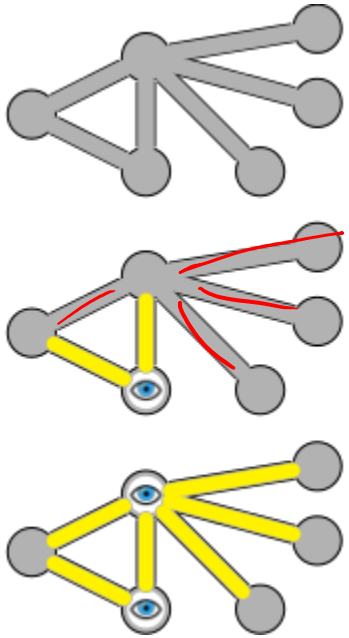
Rich
Pure Graph Theory

Sensitivity: Internal

Vertex Cover

A set of vertices that includes at least one endpoint of every edge





not

is a vertex cover

Sensitivity: Internal

Minimum Vertex Cover





Node variables x_u

Edge variables x_{uv}

$x_1 = \begin{cases} 1 \rightarrow \text{Pick vertex 1} \\ 0 \rightarrow \text{don't} \end{cases}$

Labels ℓ_v

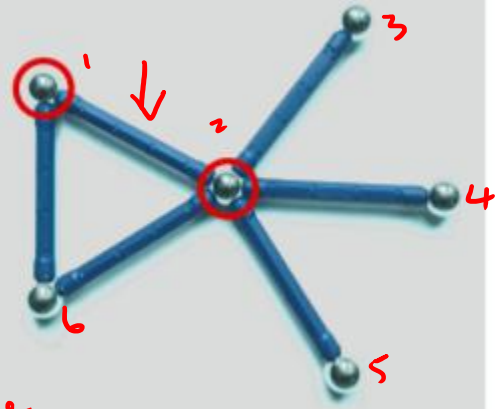
Example 2: Min. vertex cover

$$\min \sum_{u \in V} x_u$$

s.t.

$$\forall \{u, v\} \in E: x_u + x_v \geq 1$$

↑ repeat for every edge



V - all vertices $= \{1, 2, 3, 4, 5, 6\}$

E - all edges $= \{(1,2), (2,3), (2,4), (2,5), (2,6), (1,6)\}$

Min $x_1 + x_2 + x_3 + x_4 + x_5 + x_6$

s.t

$(1,2) \rightarrow \quad x_1 + x_2 \geq 1$

$(2,3) \rightarrow \quad x_2 + x_3 \geq 1$

Sensitivity: Internal

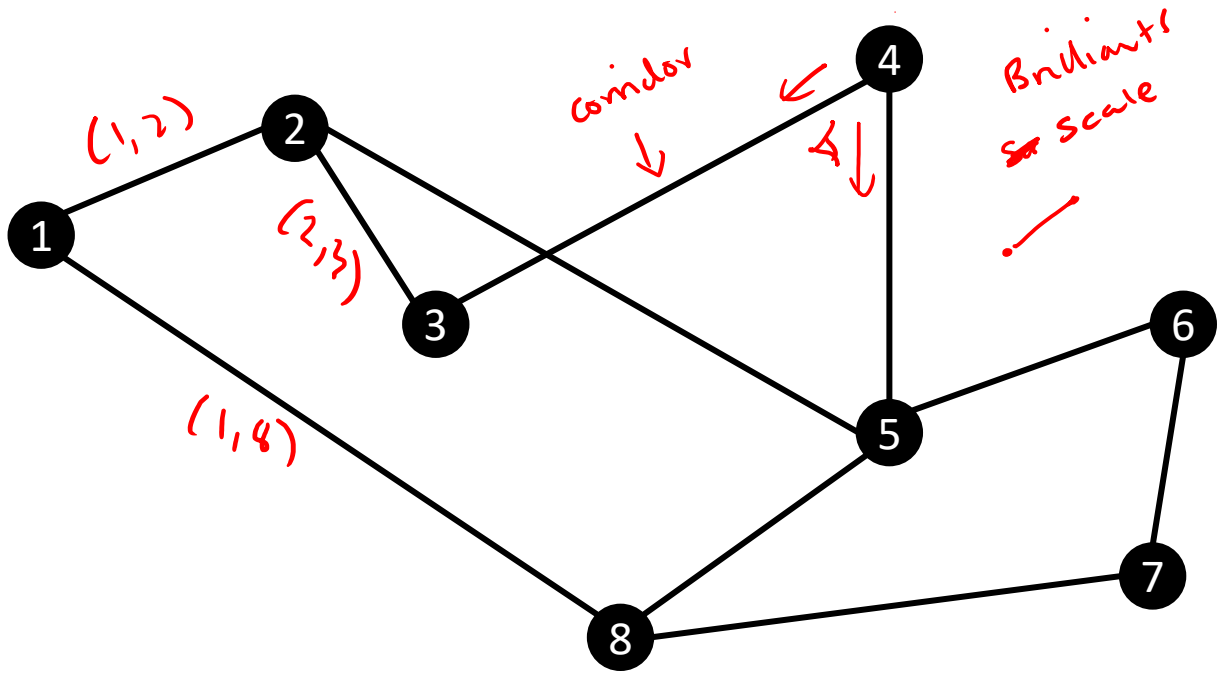
Example – Installing Security Cameras

- You wish to install security cameras in a school to cover every corridor.
- Obviously you would like to install as few as needed

- The following example is from https://brilliant.org/wiki/vertex-cover/

Sensitivity: Internal

Example – Installing Security Cameras



minimise

$$\sum_{u \in V} x_u = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8$$

subject to:

x_v - $\begin{cases} 1 & \text{camera} \\ 0 & \text{no camera} \end{cases}$

NP-Hard

$$\begin{cases} x_1 + x_2 \geq 1 & (1,2) \\ x_1 + x_8 \geq 1 & (1,8) \\ x_2 + x_3 \geq 1 & (2,3) \\ x_2 + x_5 \geq 1 \\ x_3 + x_4 \geq 1 \\ x_4 + x_5 \geq 1 \\ x_5 + x_6 \geq 1 \\ x_5 + x_8 \geq 1 \\ x_6 + x_7 \geq 1 \\ x_7 + x_8 \geq 1 \end{cases}$$

$x_u \in \{0,1\}$ ← Binary

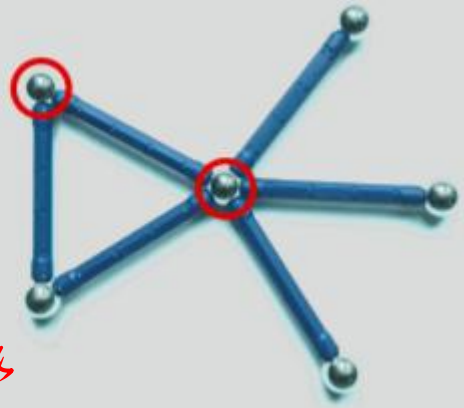
Example 2: Min. vertex cover

$$\min \sum_{u \in V} x_u$$

s.t.

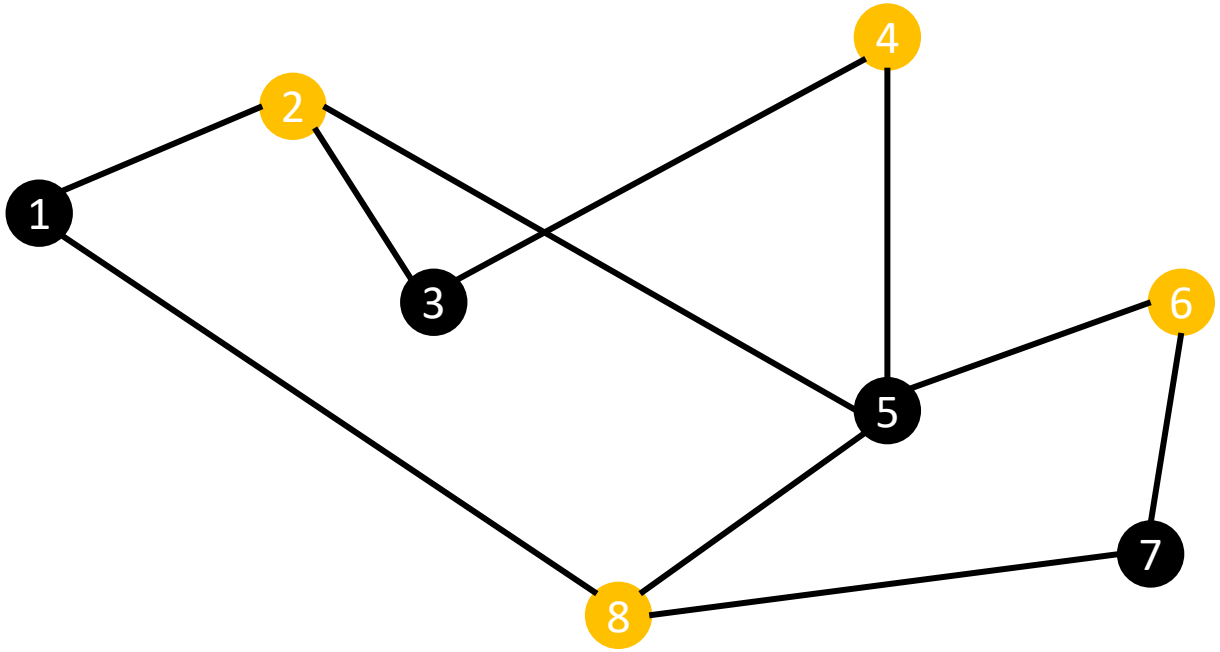
$$\forall \{u,v\} \in E: x_u + x_v \geq 1$$

min vertices to include edges



Sensitivity: Internal

Example – Installing Security Cameras



minimise

$$\sum_{u \in V} x_u = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8$$

subject to:

$$x_1 + x_2 \geq 1$$
$$x_1 + x_8 \geq 1$$
$$x_2 + x_3 \geq 1$$
$$x_2 + x_5 \geq 1$$
$$x_3 + x_4 \geq 1$$
$$x_4 + x_5 \geq 1$$
$$x_5 + x_6 \geq 1$$
$$x_5 + x_8 \geq 1$$
$$x_6 + x_7 \geq 1$$
$$x_7 + x_8 \geq 1$$

$$x_u \in \{0,1\}$$

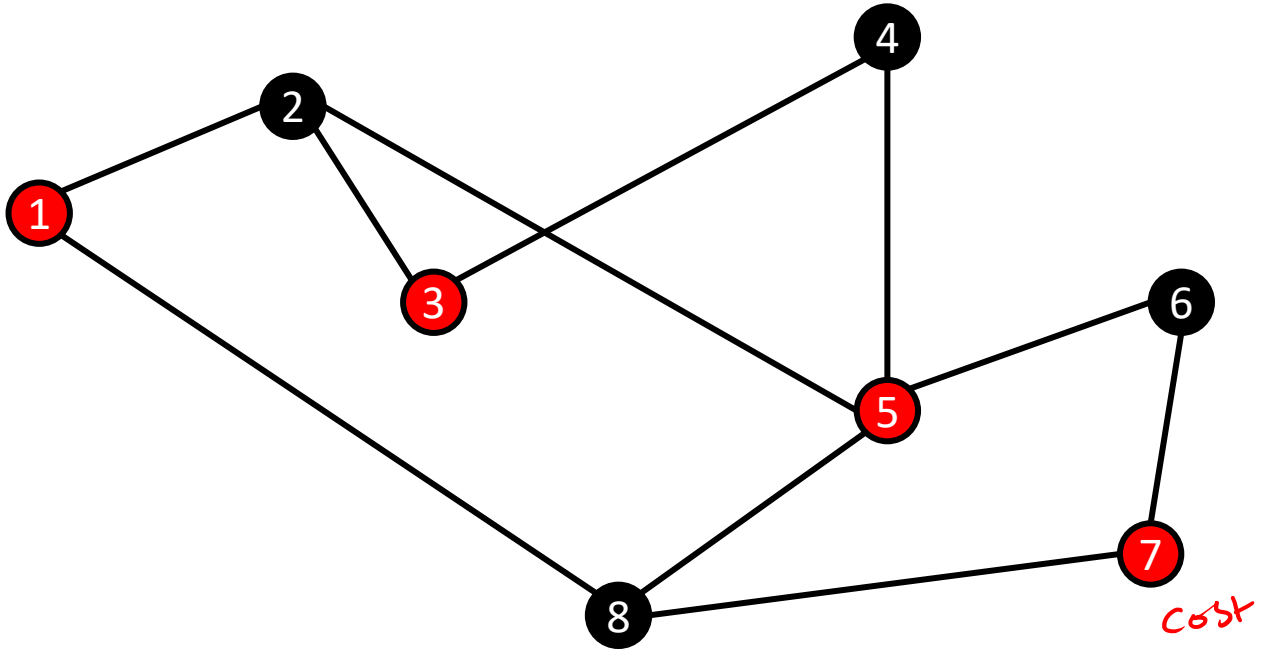
2, 4, 6, 8

objective value = 4

MATLAB Solution

Sensitivity: Internal

Example – Installing Security Cameras



Another possibility

minimise

$$\sum_{u \in V} x_u = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8$$

$c_1 x_1 + \cdots + c_8 x_4$

subject to:

$$x_1 + x_2 \geq 1$$
$$x_1 + x_8 \geq 1$$
$$x_2 + x_3 \geq 1$$
$$x_2 + x_5 \geq 1$$
$$x_3 + x_4 \geq 1$$
$$x_4 + x_5 \geq 1$$
$$x_5 + x_6 \geq 1$$
$$x_5 + x_8 \geq 1$$
$$x_6 + x_7 \geq 1$$
$$x_7 + x_8 \geq 1$$

$$x_u \in \{0,1\}$$

Sensitivity: Internal

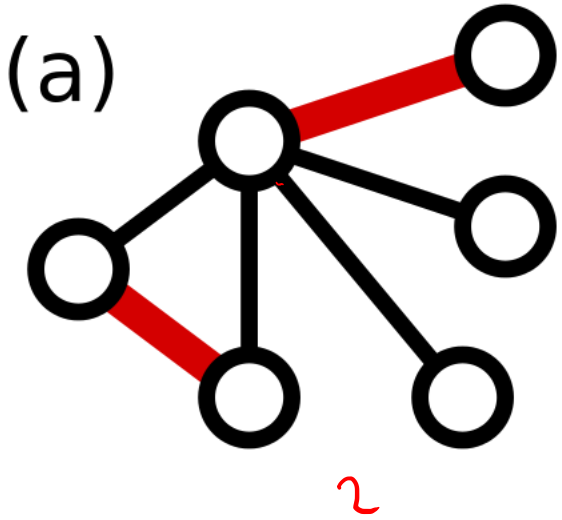
Maximum Weight Matching and Assignment Problem

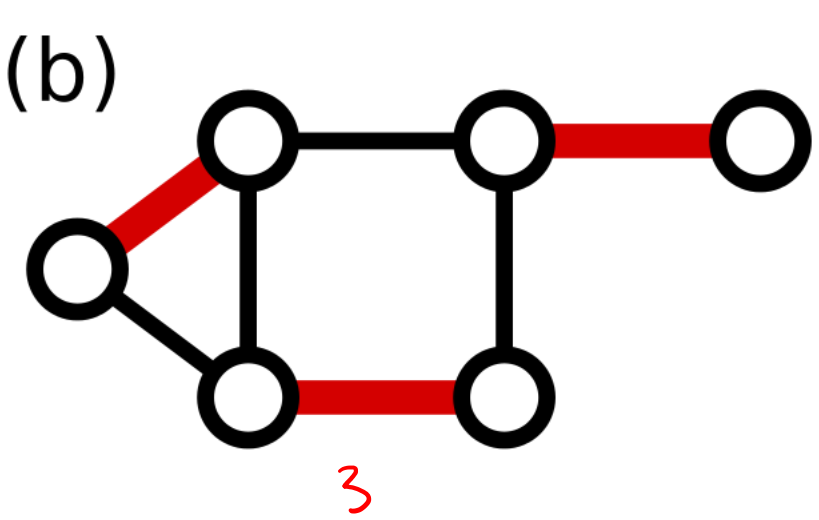
Sensitivity: Internal

Maximum Matching

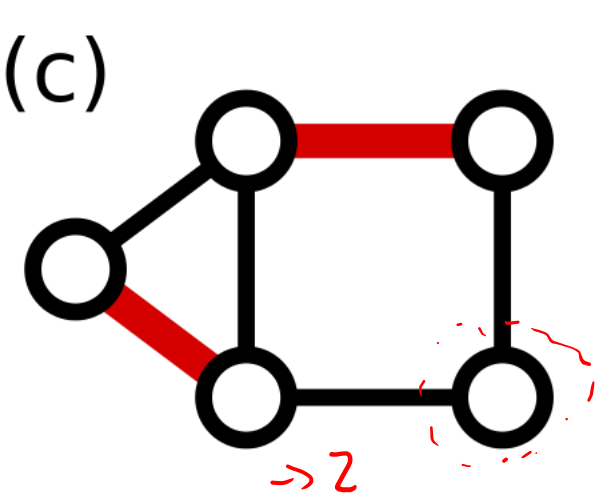
- Matching – Set of edges without common vertices
- Maximum Matching – Matching with the largest number of edges

Examples of Maximum Matchings





many solutions



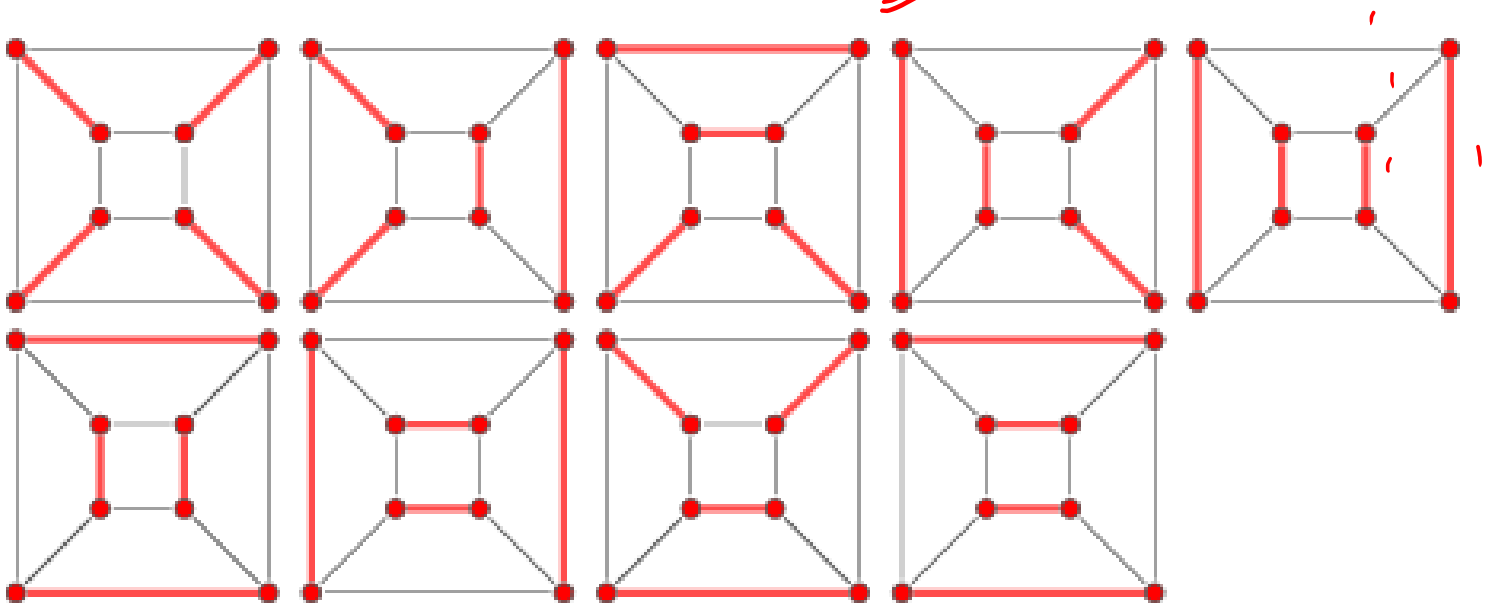
Sensitivity: Internal

Perfect Matching

- Matching – Set of edges without common vertices
- Perfect Matching – Matching that includes every vertex

Given a graph $G = (V, E)$

<u>Examples</u>



Sensitivity: Internal

Maximum Weight Matching



Node variables x_u

Edge variables x_{uv}

Labels ℓ_v

other example from Vodafone

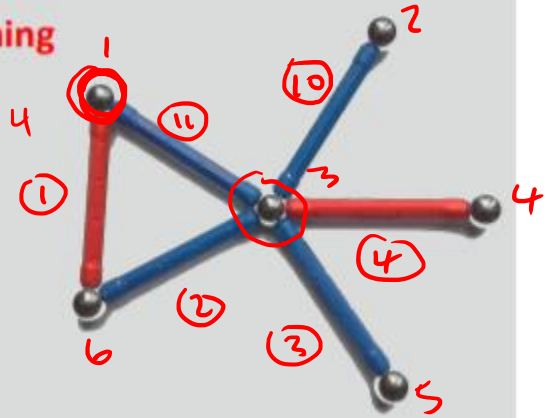
Example 1: Max. weight matching

$$\max \sum_{\{u,v\} \in E} w_{uv} \cdot x_{uv}$$

s.t.

$$\forall u \in V: \sum_{\{u,v\} \in E} x_{uv} \leq 1$$

for every vertex



- Select a set of edges without common vertices
- Maximise the weight of the edges

$x_{uv} = \begin{cases} 1 & (u,v) \text{ picked} \\ 0 & (u,v) \text{ not picked} \end{cases}$

$V = \{1, 2, 3, 4, 5, 6\}$

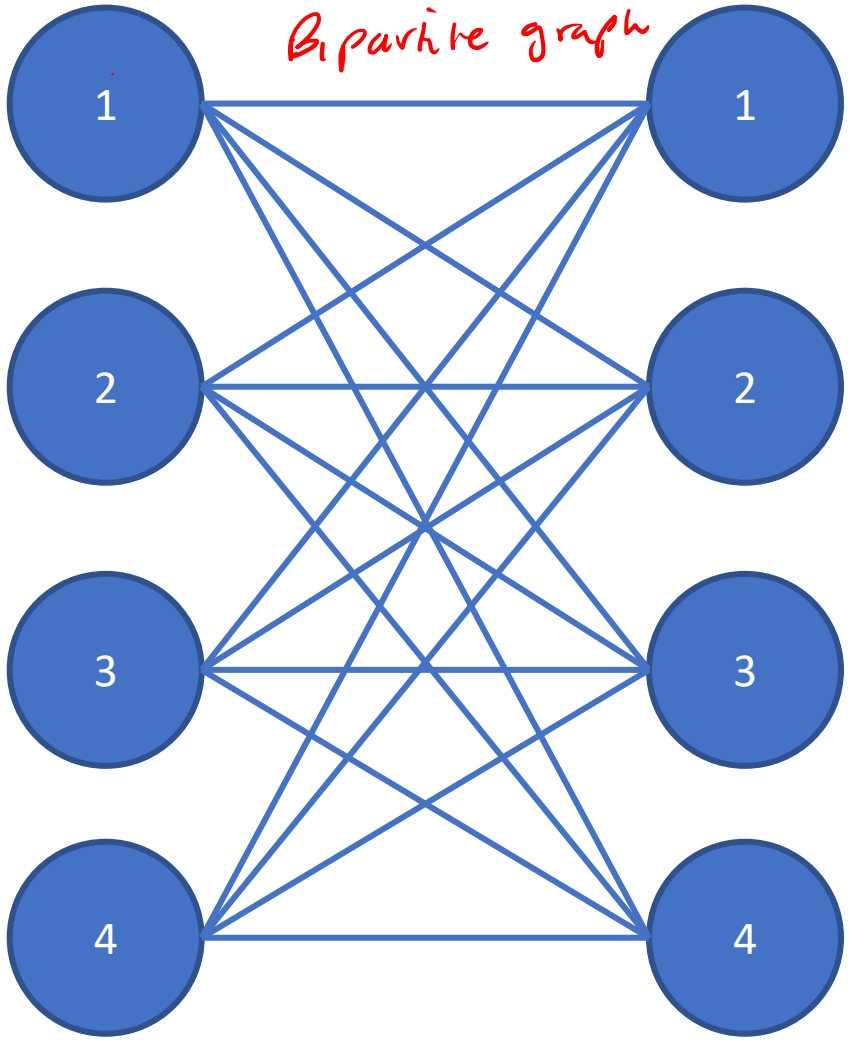
Con for vertex 1: $x_{13} + x_{16} \leq 1$

Sensitivity: Internal

Balanced Assignment Problem

- Assign a worker to each job – minimise completion time

	Job 1	Job 2	Job 3	Job 4
Worker 1	9	2	7	8
Worker 2	6	4	3	7
Worker 3	5	8	1	8
Worker 4	7	6	9	4



This is a bi-partite graph and requires a perfect matching…

How many solutions??

4!

In general $n!$

Sensitivity: Internal

Balanced Assignment Problem

- Example Solution

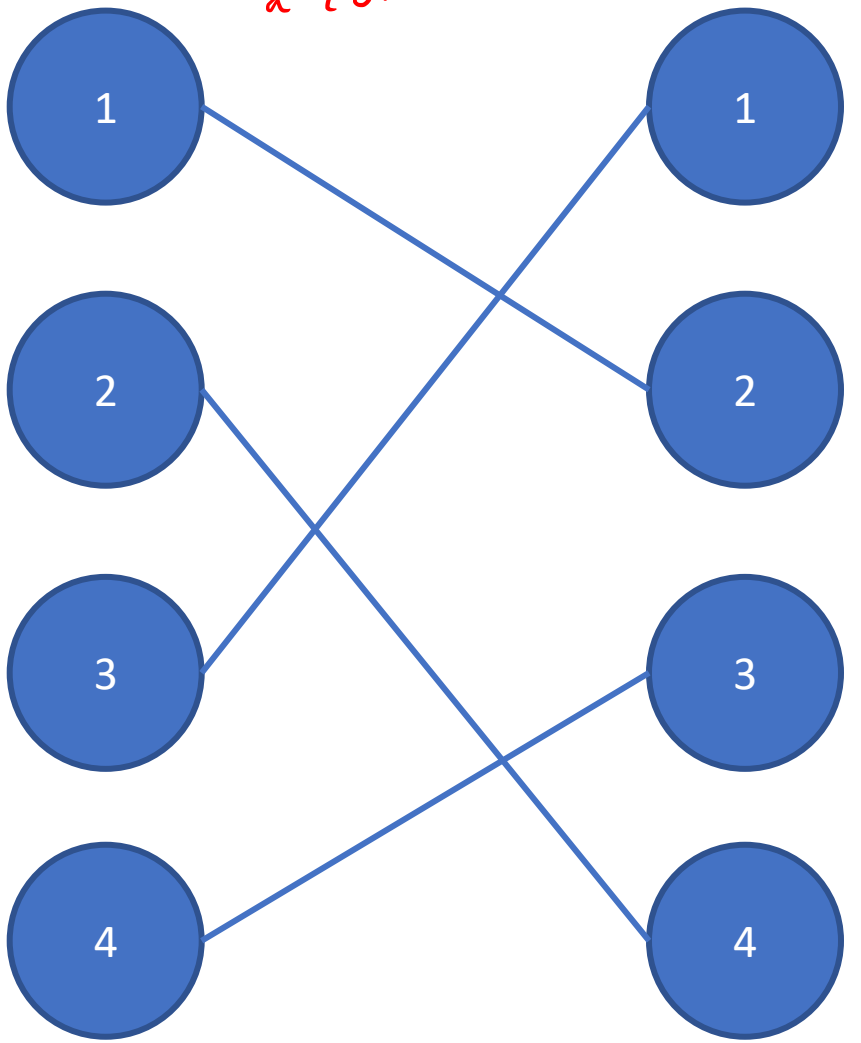
	Job 1	Job 2	Job 3	Job 4
Worker 1	9	2	7	8
Worker 2	6	4	3	7
Worker 3	5	8	1	8
Worker 4	7	6	9	4

This is a bi-partite graph and requires a perfect matching…

Here every vertex is included and no edge has a common vertex

Cost is $2 + 7 + 5 + 9 = 23$ ✓

Included every vertex every edge doesn't have a common vertex



Sensitivity: Internal

Balanced Assignment Problem – Model

- Pick 1 Per Row and 1 Per column

$x_{uv} = \begin{cases} 1 & u \text{ completes } v \\ 0 & \text{otherwise} \end{cases}$

	Job 1	Job 2	Job 3	Job 4
Worker 1	9	2	7	8
Worker 2	6	4	3	7
Worker 3	5	8	1	8
Worker 4	7	6	9	4

W - Set of Workers
J - Set of Jobs
x_{uv} - Worker u completes job v
c_{uv} - Completion time for worker u completing job v

minimise

$$\sum_{u \in W} \sum_{v \in J} c_{uv} x_{uv} = c_{11}x_{11} + c_{12}x_{12} + \cdots + c_{uv}x_{uv} + \cdots + c_{44}x_{44}$$

subject to:

$$\sum_{v \in j} x_{uv} = 1, \quad \forall u \in W$$

$$\sum_{u \in W} x_{uv} = 1, \quad \forall v \in J$$

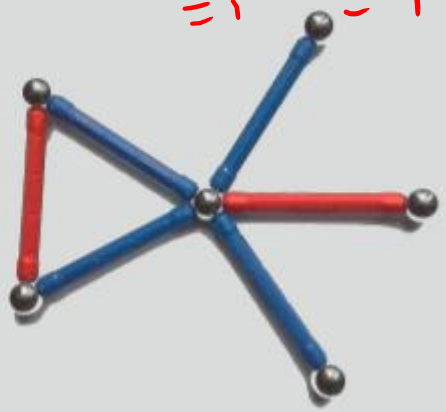
$$x_{uv} \in \{0,1\}$$

Example 1: Max. weight matching

$$\max \sum_{\{u,v\} \in E} w_{uv} \cdot x_{uv}$$

s.t.

$$\forall u \in V: \sum_{\{u,v\} \in E} x_{uv} \leq 1$$



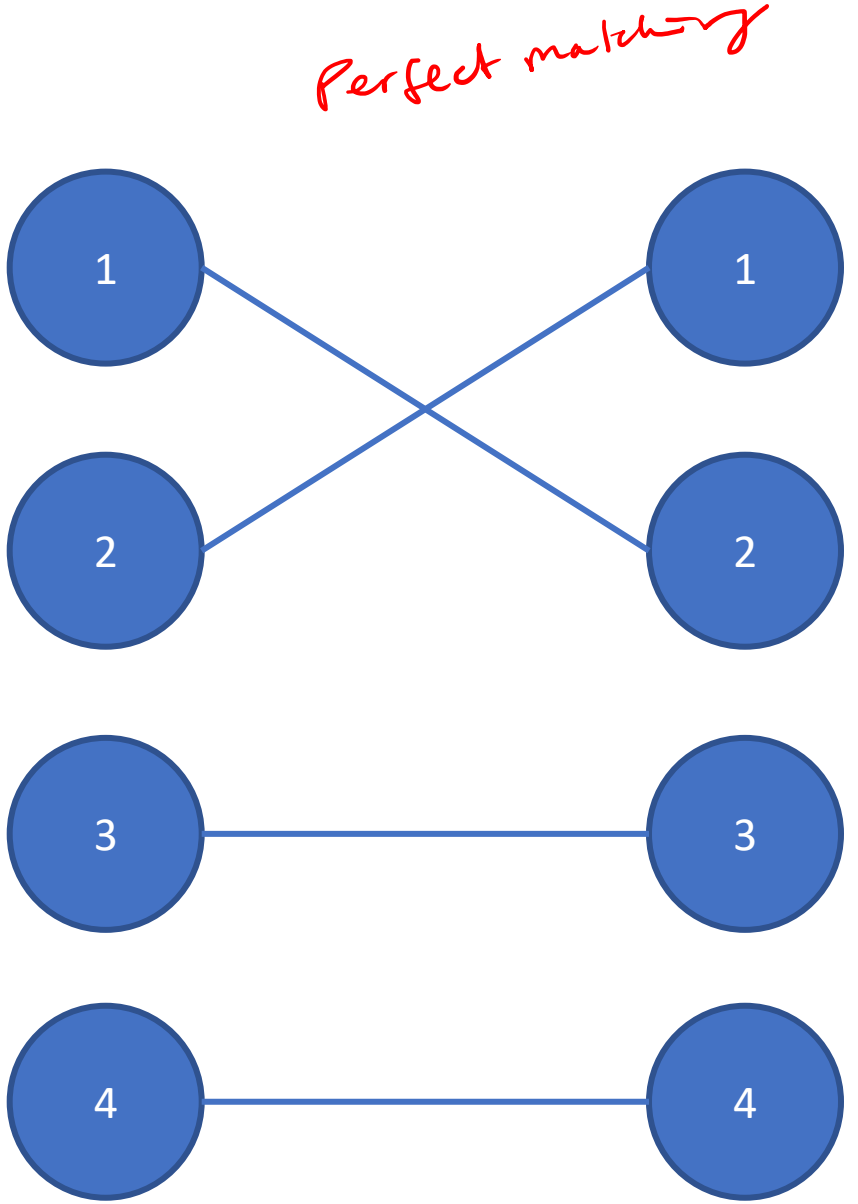
Sensitivity: Internal

Balanced Assignment Problem

- Optimal Solution

$x_{12} = 1$ $x_{21} = 1$ $x_{33} = 1$ $x_{44} = 1$

	Job 1	Job 2	Job 3	Job 4
Worker 1	9	2	7	8
Worker 2	6	4	3	7
Worker 3	5	8	1	8
Worker 4	7	6	9	4



This is a bi-partite graph and requires a perfect matching…

Here every vertex is included and no edge has a common vertex

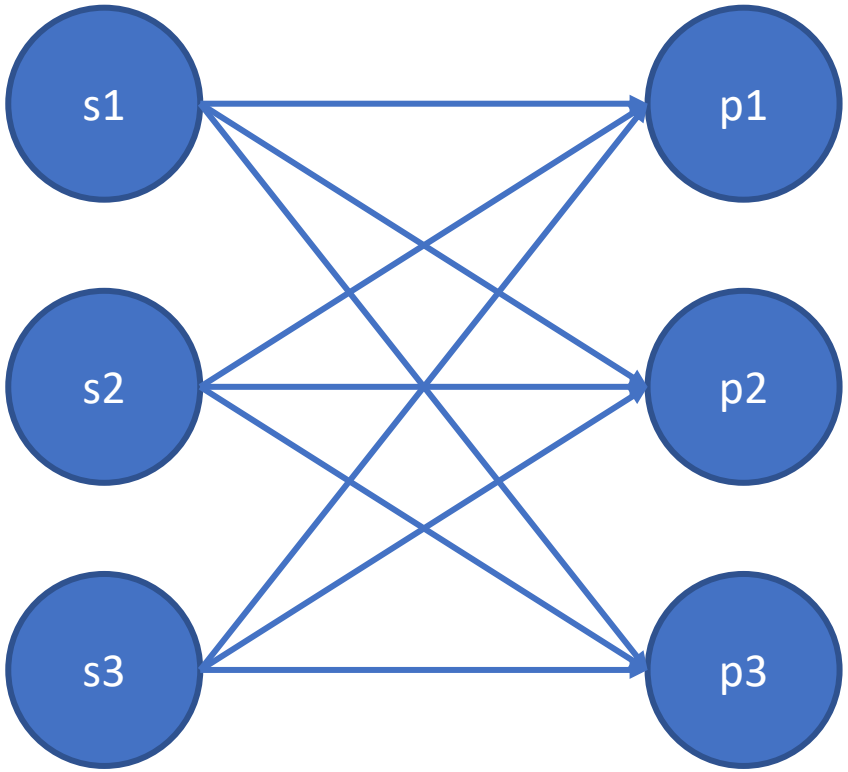
Cost is $2 + 6 + 1 + 4 = 13$

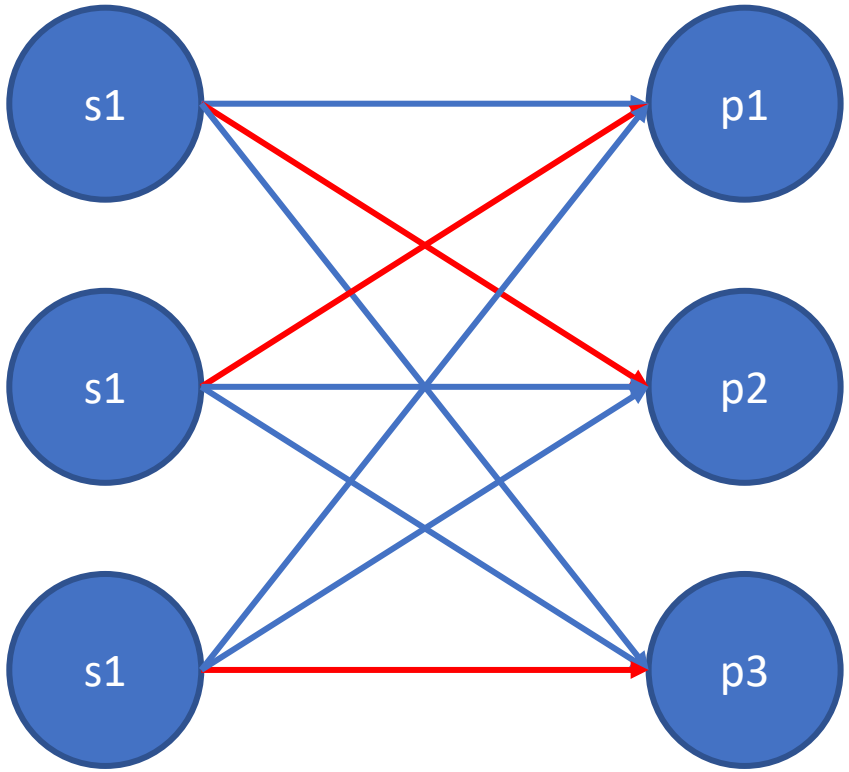
Sensitivity: Internal

Assignment Problem

$$X = \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \\ x_{41} & x_{42} & x_{43} & x_{44} \end{bmatrix}$$

Assign 3 students to the 3 different dissertation projects.





Sensitivity: Internal

Homework

Study these models and understand the MATLAB code. It will be useful for coming lectures.

Try coding these yourself.

Look up other problems such as.

- Maximum Set Cover
- Maximum Edge Cover
- Maximum Independent Set

- Look at the examples on Gurobi

https://www.gurobi.com/resource/modeling-examples-using-the-gurobi-python-api-in-jupyter-notebook/

A lot of these are taken from ‘Model Building in Mathematical Programming’, by H. Paul Williams

Sensitivity: Internal